SAHAYAK HEALTH AI

Final Capstone Report

Agentic decision-support assistant for community health workers

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Report summary	Description
Objective	Convert a first-person symptom description into exactly one care-seeking recommendation: WAIT, DOCTOR, or ER, without diagnosing or prescribing.
Architecture	Two-phase sequential agent workflow with a human-in-the-loop follow-up pause, deterministic safety auditing, and artefact-backed evaluation.
Final evaluation	Stage 4 focus: Failure analysis → train-only prompt calibration → frozen held-out evaluation → dashboard and final communication.

⚠ WARNING — EDUCATIONAL PROTOTYPE

This system is an educational prototype. It must not be represented as clinically validated or used as a replacement for qualified medical judgement.

Executive Summary

- Sahayak Health AI is an agentic triage assistant designed for ASHA/community health workers. The system does not attempt to diagnose a disease. Instead, it extracts symptoms, estimates care-seeking urgency, asks one targeted follow-up question when the severity is ambiguous, and produces exactly one recommendation: WAIT, DOCTOR, or ER.
- The solution uses a two-phase sequential architecture. Phase 1 performs intake and information gathering and then pauses for the health worker's answer. Phase 2 consumes the original information plus that answer, makes the triage decision, formats a safe response, and performs a safety audit. The design explicitly follows an escalation-only principle: follow-up information may increase urgency but must never reduce the base recommendation. Stage 4 converts Solution v1 into Solution v2 by first diagnosing systematic failures, then recalibrating the severity rubric using only the training split. After calibration, the system is frozen and evaluated once on held-out test data against pre-declared safety and performance thresholds.
- The Week 4 notebook records a final evaluation on 50 locked cases (seed 42). Final ADK run reports 70.0% exact accuracy, 81.2% ER recall, 6.0% under-triage, 24.0% over-triage, zero ER case predicted as WAIT, 78.0% safety-evaluator pass rate, and a 38.0% human-review rate.
- The result is mixed. Exact accuracy exceeds the 60% target, and over-triage remains below the 50% ceiling. However, under-triage exceeds the 5% limit, ER recall is below the notebook's 95% target. The prototype therefore does not pass the stated non-negotiable safety gate.

Headline results

Metric	Final result	Target/interpretation
Exact accuracy	70.0%	Target ≥ 60% — PASS
ER recall	81.2%	Target ≥ 95% in acceptance table — FAIL
Under-triage rate	6.0%	Target ≤ 5% — FAIL (+1)
ER → WAIT	0 cases	Target 0 — PASS
Over-triage rate	24.0%	Target ≤ 50% — PASS
Safety-audit pass rate	78.0%	Report rate; not a substitute for triage safety
Human-review rate	38.0%	38% of evaluated cases flagged for review

Business Problem, Scope and Methodology

The target user is an ASHA/community health worker. The problem is to convert free-text symptom descriptions into a consistent care-seeking recommendation while avoiding diagnosis and prescribing. Under-triage is treated as the primary safety concern because it can assign a lower level of care than the reference label

Element	Definition in this project
Intended user	ASHA / community health worker
Input	Natural-language symptom description and available context
Allowed output	WAIT, DOCTOR, or ER
Non-goals	Disease diagnosis, prescriptions, or treatment plans
Urgency order	WAIT < DOCTOR < ER
Evaluation run	50 locked cases, seed 42, from the Week 4 notebook
Evaluation discipline	Calibration uses a separate stratified 60-case training sample; the locked evaluation should not be used for prompt tuning
Primary safety concern	Under-triage is more harmful than over-triage

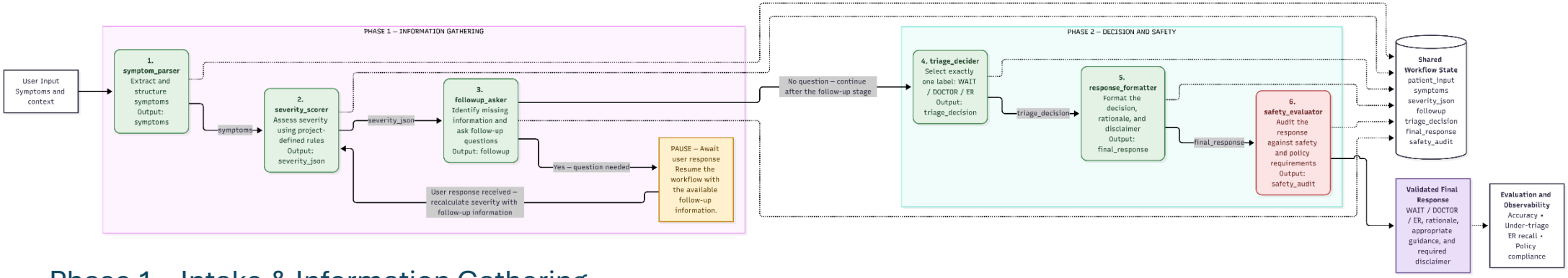
Method

- Build a sequential agent pipeline and inspect the intermediate state outputs.
- Review failure cases and group recurring issues by the stage where they originate.
- Compare the original and calibrated severity scorer on a stratified 60-case training sample.
- Run the locked 50-case evaluation and apply the deterministic safety harness.
- Report triage metrics, safety audit results, and representative demonstrations.

Measurement limitation: the evaluation set is small (n=50), so a small number of cases can change percentages materially. The ground-truth triage labels are the notebook's reference labels; they should not be interpreted as proof of clinical correctness.

Solution Architecture

The architecture separates information gathering from final decision-making. Each agent has one job and writes a designated state key, which makes intermediate outputs inspectable and supports debugging.(Used **Mermaid.ai** for Architecture visualization)



Phase 1 – Intake & Information Gathering

Stage	Input → output	Responsibility
1. Symptom Parser	Raw narrative → symptoms JSON	Extract reported symptoms and relevant context.
2. Severity Scorer	Symptoms → severity 1–5 + reason	Apply calibrated urgency rubric and consider red flags/functional impact.
3. Follow-up Asker	Severity/context → question or no question	Ask only for ambiguous severity 2–3; target missing observable information.

Human-in-the-loop pause

- **Step 1: Phase 1 generates a follow-up question**: The system processes the patient's initial symptoms and determines whether additional information is needed. It generates a follow-up question for the health worker to ask.
- **Step 2: Workflow pauses**: The automated workflow stops at the human interaction point. Phase 2 does not proceed until the required follow-up answer is available.
- **Step 3: Health worker collects the answer**: The health worker asks the patient the generated question and records the patient's response. The response may clarify the presence, absence, or progression of a symptom.
- **Step 4: Answer is stored in session state**: The recorded answer is saved in the workflow's session state, typically under the follow-up key. This preserves the response so subsequent agents can access it.
- **Step 5: Phase 2 resumes**: The saved answer is passed to Phase 2 alongside the original patient input, extracted symptoms, and severity result. The triage decision agent uses this information to select WAIT, DOCTOR, or ER according to its rules.

Phase 2 – Decision & Safety

Stage	Input → output	Responsibility
4. Triage Decider	Symptoms + severity + follow-up → label + rule	Re-evaluate with the new information; escalate on red flags; never de-escalate.
5. Response Formatter	Decision + context → response	Action-first, calm language, exact disclaimer, no diagnosis/prescription.
6. Safety Evaluator / harness	Raw response → PASS/FAIL, violations, review flag	Check prohibited content, one-label rule, disclaimer, and safety compliance.

Tools available to the triage decider

1. **search_symptom_cases_db**: Clinical Case Retrieval: Uses hybrid search over a SQLite clinical case database to retrieve similar historical cases and their triage decisions. Returns matching evidence and a similarity-weighted consensus.
2. **lookup_drug_safety**: Medication Safety Lookup: Queries the FDA drug-label API for medication warnings, contraindications, and overdose information. Supports Indian brand-name mappings and India-specific reference data for selected drugs.
3. **parse_vitals_from_text:** Vital Signs Extraction: Uses deterministic pattern matching to extract vital signs from text, including temperature, oxygen saturation (SpO_2), heart rate, respiratory rate, systolic blood pressure, and consciousness indicators. Converts Fahrenheit temperatures to Celsius.
4. **calculate_india_news2**: Physiological Risk Assessment: Calculates a NEWS2-based score from available vital signs and applies configured India-specific context modifiers. Returns a risk category, recommended escalation level, clinical flags, and a confidence indicator based on the number of available measurements.

Safety principles

- Return one allowed triage label and preserve the decision in the formatted response.
- Avoid disease claims and medication dosage/prescription instructions.
- Include the exact disclaimer: "**This is decision support guidance only. Always consult a qualified medical professional for diagnosis and treatment**."
- Emergency evidence should not be downgraded by a follow-up response; the implemented rubric and code should be reviewed for any exceptions.

Stage 4 Calibration, Evaluation and Communication

Task 4.1 — Failure Analysis

Pattern from notebook	Affected cases	Stage	Fix
1. Severity-3 assignments lead to WAIT despite reported symptoms, including stomach pain and difficulty swallowing.	8	severity_scorer / triage_decider	Do not assign severity 1 when symptoms are present; clarify rubric handling for symptoms not covered by explicit categories.
2. Breathing difficulty with multiple concerning symptoms is assigned DOCTOR instead of the reference ER label.	5	severity_scorer / triage_decider	Review emergency escalation criteria and ensure combinations of symptoms are assessed before selecting the final label.
3. Vague vomiting frequency is interpreted as persistent vomiting, causing DOCTOR predictions for reference-WAIT cases.	2	severity_scorer	Require evidence for persistence; do not infer unreported frequency or duration.

Task 4.2 — Calibrate Prompts on the Train-Split

Agent_modified: **severity_scorer**

Change_made:

Revised the severity scoring rules to classify severity 1 only when the patient explicitly reports no active symptoms.
Added a severity-3 fallback for reported symptoms that do not clearly match the rubric. (*These WAIT → DOCTOR transitions increase the likelihood of a DOCTOR prediction, serving as a precautionary measure when symptoms are not clear.*)
Expanded red-flag criteria to include severe breathing difficulty, clarified symptom matching using plain-language descriptions and synonyms, and required the highest applicable severity rule to be used.
Retained restrictions against diagnosis and invented symptoms.

Reason:

Failure analysis identified symptoms being assigned severity 1 despite being present, as well as insufficient recognition of symptom descriptions.
The changes aim to reduce under-triage and improve rubric consistency.

A/B Testing metrics Comparison

Metric	V1	V2 calibrated rubric	Improvement
Exact accuracy	58.3%	68.3%	+10.0 percentage points
Under-triage rate	26.7%	8.3%	−18.4 percentage points
ER recall	60.0%	75.0%	+15.0 percentage points

A/B Testing Protocol and Execution

- **Objective:** Evaluate whether calibrating the severity scorer improves triage classification before the final held-out evaluation.
- **Dataset:** Created a stratified 60-case sample using load_sahayak_dataset() and random_state=42, with 20 cases each for WAIT, DOCTOR, and ER.
- **Arm A — Baseline:** The original severity rubric was evaluated using run_adk_evaluation().
- **Arm B — Calibrated:** The same dataset was evaluated using severity_scorer_v2, which clarified severity criteria, added a severity-3 fallback for uncovered symptoms, prioritized the highest applicable rule, improved plain-language matching, and prohibited invented symptoms. Other agent instructions remained unchanged.
- **Metrics:** get_ab_metrics() calculated exact accuracy, under-triage rate, and ER recall after excluding records with missing labels.
- **Conclusion:** Arm B improved all three metrics but did not meet the under-triage (≤5%) or ER-recall (>95%) targets. The separate locked 50-case evaluation was retained for final assessment and was not used for prompt tuning. These calibration results indicate improvement, not clinical validation.

Model Comparison and Selection

- The executed 60-case calibration A/B comparison favours V2 on all three recorded metrics: 68.3% accuracy, 8.3% under-triage, and 75.0% ER recall, compared with V1 at 58.3%, 26.7%, and 60.0%, respectively. This supports retaining V2 for the final run, while recognizing that the calibration under-triage and ER-recall goals were not met.
- V2 improved the calibration metrics relative to V1, but these results do not establish clinical safety. The final locked evaluation must be assessed separately and remains below the ER-recall and under-triage thresholds.
- V2 was retained for the final evaluation because the executed calibration A/B results were better than V1 on the three reported metrics. However, the final locked evaluation still fails the ER-recall and under-triage targets, so these results do not establish clinical safety or readiness for deployment.

Task 4.3 — Final Held-Out Evaluation

The final raw evaluation cell contains 50 cases and reports the metrics below.

Reference \ Predicted	WAIT	DOCTOR	ER	Total
WAIT	6	11	0	17
DOCTOR	0	16	1	17
ER	0	3	13	16
Total	6	30	14	50

The matrix is read using the ordered care levels WAIT < DOCTOR < ER. The most safety-critical cells are ER→WAIT and ER→DOCTOR because they represent under-triage.

Final Results

Metric	Policy evaluation	V1 — Baseline	V2 — Calibrated (Final result)	Threshold / status
Exact accuracy	55.0%	64.0%	70.0%	≥ 60% — PASS
ER recall	0.0%	62.5%	81.2%	≥ 95% — FAIL
Under-triage rate	46.0%	28.0%	6.0%	≤ 5% — FAIL (+1%)
ER → WAIT	-	3	0	0 — PASS
Over-triage rate	-	-	24.0%	≤ 50% — PASS
Safety-audit pass rate	54.0%	58.0%	78.0%	Reported
Human-review rate	-	30.0%	38.0%	Reported

Task 4.4 — Demonstrations and Communication

The notebook includes three illustrative demonstrations that expose intermediate state and final output. They demonstrate the intended interaction pattern, but they are examples rather than evidence of overall safety.

Demo	Input summary	Recorded output	What it demonstrates
WAIT	Mild forearm itching for two days; no fever	WAIT	Low-severity example with symptom summary and return precautions.
DOCTOR	Dizziness, nausea, headache for three days	DOCTOR	Moderate symptom narrative; decision rule cites severity 3 with follow-up unavailable.
ER	Chest pain, breathing difficulty, left-arm numbness, sweating	ER	Explicit red-flag narrative; severity 5 and emergency escalation.

The follow-up/medicine-note example records a first-pass and final decision of DOCTOR and does not ask a follow-up question for that case. The response relays medication context but should not be described as a validated medication recommendation.

Safety-gate conclusion

- Every final response should contain exactly one care level, calm action-oriented language, the required disclaimer, and no diagnosis or prescription. The system is a decision-support prototype and must not present itself as a medical device.
- Using the final held-out evaluation cell, the safety gate FAILS: under-triage is 6.0%, above the 5% threshold, and ER recall is 81.2%, below the 95% target. ER-sent-home is zero and passes its target.
- The final raw evaluation's safety harness passed 78.0% of outputs, and 38.0% were flagged for human review.
- When symptoms are unclear, the model often predicts DOCTOR for cases labeled WAIT (11) and ER (3), suggesting a cautious tendency to recommend medical attention.

Limitations and Recommendations

Sahayak Health AI is an educational decision-support prototype. The limitations below constrain how its evaluation results should be interpreted and what is required before any real-world use.

Limitation / risk	Implication	Recommendation
Model ceiling	Symptom text cannot perfectly determine the patient's true clinical state.	Treat metrics as decision-support evidence, not diagnostic truth. Prioritize under-triage and catastrophic errors over accuracy alone.
Proxy ground truth	Diagnosis-to-triage mapping is an evaluation construct and may contain ambiguity or label noise.	Document the mapping assumptions and inspect edge cases against clinically reviewed guidance.
Dataset size	The 853 training / 212 test split is small for broad clinical deployment.	Evaluate on larger, representative, clinically reviewed datasets before real-world use.
Non-medical-device status	The prototype is not a certified clinical device and has not been established as clinically validated.	Maintain human oversight and obtain appropriate clinical and regulatory review before any deployment.
LLM variability and safety asymmetry	Model settings can change outputs, and under-triage can be substantially more harmful than over-triage.	Use deterministic configuration and concise, task-specific prompts where possible. Track safety metrics and preserve saved evaluation artifacts.

These limitations mean the reported metrics should guide iterative testing, not be interpreted as evidence of clinical readiness. The locked evaluation set should remain separate from prompt calibration.

Conclusion and Next Steps

- Sahayak Health AI demonstrates an agentic workflow for symptom extraction, severity scoring, follow-up handling, triage selection, response formatting, and post-hoc safety auditing. The final 50-case evaluation reports 70.0% accuracy, 81.2% ER recall, 6.0% under-triage, zero ER-sent-home cases, a 78.0% safety-harness pass rate, and a 38.0% human-review rate.
- The system passes the accuracy and ER-sent-home targets but fails the ER-recall and under-triage targets.
- Before any deployment consideration, review the three reference-ER cases not predicted as ER, investigate over-triage and missing-follow-up behavior, and rerun the notebook's reporting cells in sequence to replace stale intermediate outputs. Recompute the confusion matrix directly from the **final_adk_results** produced by the final cell. Calibration should use training data; the locked evaluation must not be used to tune prompts. The safety harness is post-hoc in this notebook, so do not claim it blocks unsafe outputs unless that behavior is implemented and verified.
- Suggest to use more advance models like **Gemini 3** or **GPT 5.6** for more robust results.

Deployment - Interactive Demo App

The deployment files package Sahayak as an interactive web demo. The app is served by FastAPI and Uvicorn, while the user interface is delivered as HTML/JavaScript. The demo exposes the main interaction at '**Simple demo**' page and a trust dashboard at '**dashboard**' page.

Application startup and configuration

Component	Purpose
FastAPI + Uvicorn	Serves the web application
HTML/JavaScript interface	Provides the Simple demo and Trust dashboard
Ollama Cloud API	Runs the configured language model
.env file	Stores the API key and model configuration
Local case database	Supports historical-case retrieval
Safety and evaluation tools	Check outputs and calculate benchmark metrics

Configure environment variables

- Create a local **.env** file for development. For example:

```
OLLAMA_API_KEY=your_actual_api_key
OLLAMA_API_BASE=https://ollama.com/v1
SAHAYAK_OLLAMA_MODEL=gpt-oss:120b
```

- **requirements.txt**
  - google-adk>=2.0.0, google-genai, litellm
  - datasets: pandas, numpy, matplotlib, seaborn, scikit-learn, requests
  - fastapi, uvicorn
  - pytest, ipykernel, nbclient, nbformat, jupyter, python-dotenv
- **Test locally in a clean virtual environment:**

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
python demo_app.py
```

Dashboard and evaluation support

The `**/dashboard**` page provides a 20-case benchmark, a custom-query path, a deterministic safety audit, a confusion matrix, and an optional DSPy panel. The deterministic evaluator is the reported safety judge; the LLM auditor is displayed separately, with agreement treated as a separate metric.

